

Rubén Izquierdo López

PhD Student in Mathematics

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Location: Madrid, Spain

Nationality: Spanish

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Biography

PhD student at ICMAT and Universidad Autónoma de Madrid, supported by an FPU2024 fellowship and supervised by Manuel de León. My research focuses on differential geometry, geometric mechanics, and the geometric formulation of classical field theories.

Education

- **PhD in Mathematics** (2024–present)
ICMAT and Universidad Autónoma de Madrid
FPU2024 fellow; supervised by Manuel de León.
- **Master's Degree in Advanced Mathematics** (2023–2024)
Universidad Complutense de Madrid
Average grade: 9.77/10. Master's thesis: Coisotropic Reduction in Multisymplectic Geometry.
- **Bachelor's Degree in Mathematics** (2019–2023)
Universidad Complutense de Madrid
Average grade: 9.44/10. Bachelor's thesis: Homología Singular y Celular.

Research Experience

- **PhD Researcher** (2024–present)
Instituto de Ciencias Matemáticas (ICMAT)
For developing by PhD in CSIC (Instituto de Ciencias Matemáticas) under the supervision of Manuel de León. We are currently researching 'graded Dirac' and 'graded Poisson' structures for the study of classical field theories.
- **Severo Ochoa contract for Master students** (2023–2024)
Instituto de Ciencias Matemáticas (ICMAT)
Contract for developing my master thesis under the supervision of Manuel de León. We developed the multisymplectic coisotropic reduction.
- **JAE Intro research grant** (2023)
Instituto de Ciencias Matemáticas (ICMAT)
Introduction to research grant where I learned about the world of differential geometry and geometric mechanics.

Grants and Awards

- **FPU2024 Fellowship** (2025)
Spanish Ministry of Science, Innovation and Universities
Predoctoral fellowship supporting doctoral research.
- **Bachelor's Degree Extraordinary Award** (2024)
Universidad Complutense de Madrid
Award granted for the best curriculum in the degree of Mathematics.
- **Severo Ochoa grant** (2023–2024)

Instituto de Ciencias Matemáticas (ICMAT)

Grant for Master students

- **Excelence grant** (2020-2021, 2021-2022, 2022-2023)
Comunidad de Madrid
Merit-based scholarship for undergrad students
- **JAE Intro research grant** (2023)
Consejo Superior de Investigaciones Científicas (CSIC)
Grant for introduction to research

Publications

- [1] M. de León and R. Izquierdo-López, "A description of classical field equations using extensions of graded Poisson brackets", *International Journal of Geometric Methods in Modern Physics*, Accepted for publication. [[10.1142/S0219887826502439](https://arxiv.org/abs/10.1142/S0219887826502439)]
- [2] M. de León, R. Izquierdo-López and X. Rivas, "Brackets in Multicontact Geometry and Multisymplectization", *Mediterr. J. Math.* 23, 81 (2026). [[10.1007/s00009-026-03077-4](https://arxiv.org/abs/10.1007/s00009-026-03077-4)]
- [3] M. de León and R. Izquierdo-López, "Graded poisson and graded dirac structures", *J. Math. Phys.* 66 (2), 022901 (2025). [[10.1063/5.0243128](https://arxiv.org/abs/10.1063/5.0243128)]
- [4] M. de León and R. Izquierdo-López, "Coisotropic reduction in multisymplectic geometry", *Geometric Mechanics*, 1 (3), 209–244 (2024). [[10.1142/S2972458924500096](https://arxiv.org/abs/10.1142/S2972458924500096)]
- [5] M. de León and R. Izquierdo-López, "A review on coisotropic reduction in symplectic, cosymplectic, contact and co-contact Hamiltonian systems", *J. Phys. A: Math. Theor.* 57, 163001 (2024). [[10.1088/1751-8121/ad37b2](https://arxiv.org/abs/10.1088/1751-8121/ad37b2)]

Preprints

- [1] M. de León, R. Izquierdo-López, M. Lainz, "The Spencer cohomology and integrability of multisymplectic structures". [[arXiv:2607.05267](https://arxiv.org/abs/2607.05267)]
- [2] M. de León, X. Gràcia, R. Izquierdo-López, A. Martínez-Muñoz, X. Rivas, "Poisson and Jacobi structures from 2-covariant tensors". [[arXiv:2606.20030](https://arxiv.org/abs/2606.20030)]
- [3] M. de León, R. Izquierdo-López, L. Schiavone, P. Soto, "Regularization of singular time-dependent Lagrangian systems". [[arXiv:2603.24308](https://arxiv.org/abs/2603.24308)]
- [4] M. de León, R. Izquierdo-López, L. Schiavone, P. Soto, "A zoo of coisotropic embeddings". [[arXiv:2509.19039](https://arxiv.org/abs/2509.19039)]

Seminars and Conferences

Invited Talks at Conferences

- **The Jacobi brackets of dissipative field theories** (13-17 July, 2026)
2nd International Workshop "New Advances in Contact and Symplectic Hamiltonian Dynamics"
- **Graded Brackets in Classical Field Theory: Conservative and Dissipative** (9-13 September, 2025)
XXXIII International Fall Workshop on Geometry and Physics
- **Recent Developments in the Theory of Brackets in Classical Field Theories** (16-18 June, 2025)
Geometry and Topology for the future III
- **Coisotropic reduction in Classical Field Theories** (23 September, 2024)
Workshop de Jóvenes Investigadores, UCM
- **The Multisymplectic Framework of Field Theories** (21-23 August, 2024)
Workshop on Geometric Aspects of Material Modelling

Contributed Talks at Conferences

- **Multisymplectic structures as G-structures** (6-8 July, 2026)
2nd International Workshop on Geometric Field Theory
- **On the theory of G-structures applied to multisymplectic manifolds** (22-25 June, 2026)

OAK2: Observables, Actions and Quantization

- **The graded Poisson bracket of general conservation laws in classical field theories** (19 January, 2026)
VIII Iberoamerican Meeting on Geometry, Dynamics, and Field Theory
- **Extensions of brackets in classical field theories** (23-27 June, 2025)
Summer School of Geometry, Dynamics and Field Theory 2025
- **Graded structures of classical field theory** (20 January, 2025)
XIX Young Researchers Workshop in Geometry, Dynamics and Field Theory
- **A graded approach to brackets in classical field theory** (17 January, 2025)
VII Congreso de Jóvenes Investigadores de la RSME
- **Poisson and Dirac structures in Classical Field Theories** (6 November, 2024)
New Advances in Contact and Symplectic Hamiltonian Dynamics
- **Coisotropic reduction in Multisymplectic Geometry** (13-15 June, 2024)
The Geometry of Field Theories
- **Coisotropic reduction in different phase spaces** (21-23 February, 2024)
XVIII International Young researchers Workshop in Geometry, Dynamics and Field Theory
- **Coisotropic reduction in Symplectic, Cosymplectic, Contact, and Cocontact geometry** (13-16 November, 2023)
BYMAT (Bringing Young Mathematicians Together)

Seminars

- **The graded Jacobi bracket in dissipative field theories** (06 March, 2026)
Gamma (Geometry and Applications: Modern Mathematical Approaches) Seminar
- **Natural operations in differential geometry** (7 May, 2025)
Coloquio Junior de matemáticas
- **Graded Poisson brackets in classical field theories** (5 February, 2025)
Q-Math Seminar
- **Geometría Multisimpléctica, el marco para Teorías Clásicas de Campos** (19 September, 2024)
Seminario de doctorandos, UCM
- **Coisotropic reduction in Multisymplectic Geometry** (26 June, 2024)
Gamma (Geometry and Applications: Modern Mathematical Approaches) Seminar

Posters

- **Higher form symmetries and general conservation laws** (27-31 July, 2026)
Higher Categories for the Working Mathematician
- **Higher form symmetries and general conservation laws** (25-27 March, 2026)
XX International Young Researchers Workshop on Geometry, Dynamics and Field Theory

Other Scientific Meetings I Attended

- **XVI International ICMAT Summer School on Geometry, Dynamics and Field theory** (June 2024)
Instituto de Ciencias Matemáticas (ICMAT)
- **XXV WinterMeeting on Geometry, Dynamics and Field Theory** (January 2024)
Universidad de Zaragoza
- **de León fest** (December 2023)
Instituto de Ciencias Matemáticas (ICMAT)
- **JAЕ Summer School** (July 2023)
Instituto de Ciencias Matemáticas (ICMAT)

Teaching

2026-27

- **Topología, grupo 722 (Topology)**
Grado en Matemáticas UAM

2025-26

- **Geometría lineal, grupo 711 (Linear Algebra)**
Grado en Matemáticas UAM
- **Geometría lineal, grupo 220 (Linear Algebra)**
Grado en Ingeniería Informática y Matemáticas UAM
- **Ecuaciones Diferenciales (Differential Equations)**
Grado en Física UNIR
- **Diseño de Algoritmos (Algorithm Design)**
Grado en Ingeniería Informática UNIR
- **Diseño de Algoritmos (Algorithm Design)**
Grado en Matemática Computacional UNIR

2024-25

- **Sistemas Dinámicos (Dynamical Systems)**
Grado en Matemática Computacional UNIR
- **Lógica Computacional (Computational Logic)**
Grado en Matemática Computacional UNIR
- **Análisis Complejo (Complex Analysis)**
Grado en Física UNIR
- **Análisis Numérico (Numerical Analysis)**
Grado en Física UNIR
- **Ecuaciones Diferenciales (Differential Equations)**
Grado en Física UNIR

2022-2023

- **Matemáticas Básicas (Basic Mathematics)**
Grado en Matemáticas UCM

2020-2022

- **Support Teacher**
Savannah SLC in Torrejón de Ardoz

Outreach and Volunteering

- **Volunteer, European Congress of Mathematics** (July 2024). Assisted with the organization of ECM 2024 in Seville.
- **Volunteer in "Concurso de Primavera"** (April 2023). Assisted with the organization of "Concurso de Primavera", a math competition for highschool students.

Academic Activities Organized

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- **Co-organizer, Geometry, Mechanics and Control Seminar** (2025–present). I co-organize the biweekly seminar of the research group in ICMAT.
 - **Co-organizer, Reading Groupoid** (2025–2026). Academic-year reading group on Lie groupoids, for more information, see the official website <https://readinggroupoid.org/>. This is a joint initiative with ICMAT–UCM–UW.

Languages

Spanish (native); English (professional working proficiency)